



SYDNEY BOYS HIGH
SCHOOL
MOORE PARK, SURRY HILLS

Yearly Examination 2008

Mathematics Extension

General Instructions

- Working time – 75 Minutes
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators may be used.
- All necessary working should be shown in every question.
- Answer in simplest exact form unless otherwise instructed.

Total Marks – 69

- All Questions may be attempted.
- Each Question is worth 23 marks, and should be handed up in a separate examination booklet.
- Full Marks may not be awarded for careless or poorly set out work.

Examiner – *A.M. Gainford*

Question 1. (23 Marks)

(a) Find the exact value of $\sec 210^\circ$. **1**

(b) Solve for x : **6**

(i) $|2 - x| \leq 3$

(ii) $x^2 - 4 < 0$

(iii) $\frac{1}{2x-6} < 1$

(c) Find the remainder when the polynomial $P(x) = x^3 - 5x - 1$ is divided by $x - 3$. **1**

(d) Give the general solution of the equation $2 \cos \left(\theta + \frac{\pi}{6} \right) + 1 = 0$. **3**

(e) Use the substitution $t = \tan\left(\frac{\theta}{2}\right)$ to solve the equation $2 \sin \theta + \cos \theta = 0$ for $-180^\circ \leq \theta \leq 180^\circ$. (Answer correct to the nearest minute.) **3**

(f) Express $\cos 3\theta$ as an expression in powers of $\cos \theta$ only. **3**

(g) Differentiate: **6**

(i) $x\sqrt{x} - \frac{1}{x}$

(ii) $(1 - x^2)^4 (1 + x^2)^4$

(iii) $\frac{1+x}{1-x}$

Question 2. (23 Marks)

(a) Express $\cos x - \sin x$ in the form $R \cos(x + \alpha)$, where $R > 0$, α is acute. **2**

(b) (i) Show that $2 \tan \theta \sec \theta = \frac{1}{1 - \sin \theta} - \frac{1}{1 + \sin \theta}$. **4**

(ii) Simplify $\frac{\sin 3x}{\sin x} - \frac{\cos 3x}{\cos x}$.

(c) If α, β, γ are the roots of the polynomial equation $3x^3 - 6x^2 + 3x + 1 = 0$, evaluate: **4**

(i) $\alpha\beta\gamma$

(ii) $\alpha\beta + \alpha\gamma + \beta\gamma$

(iii) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$

(d) Given the polynomial $P(x) = x^3 + 6x^2 - x - 30$. **3**

(i) Use the factor theorem to find a zero of the polynomial.

(ii) Express $P(x)$ as a product of three linear factors.

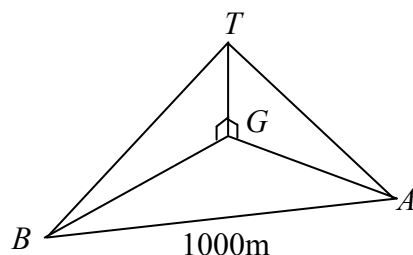
(e) Draw neat sketches of the following functions: **6**

(i) $y = 1 - |x - 1|$ (ii) $y = 2^{-x}$ (iii) $y = 1 - \sqrt{x}$

(f) Find all solutions of $\sin 2x = \sin x$ (x in radians). **4**

Question 3. (23 Marks)

- (a) Solve $\left(x + \frac{1}{x}\right)^2 - 6\left(x + \frac{1}{x}\right) + 8 = 0$. 4
- (b) Calculate to the nearest minute the acute angle between the lines $x - 3y = 6$ and $2x + y = -5$. 2
- (c) A surveyor observes, from the top (T) of a vertical cliff a kilometer-post (A) on a straight flat road on the plain below on a bearing of $176^\circ T$, and at an angle of depression of 4° . He then observes the next kilometer-post (B) (one kilometer further along the road) on a bearing of $194.5^\circ T$, and at an angle of depression of 3° . 5



- (i) Copy the diagram to your answer booklet, and find $\theta = \angle AGB$.
- (ii) Let $GT = h$ m, $AG = x$ m, $BG = y$ m, write an equation connecting x, y and θ .
- (iii) Hence find the height of the cliff.
- (d) Let $P(2ap, ap^2)$ be a point on the parabola $x^2 = 4ay$. 4
- (i) Write down the co-ordinates of the midpoint M of the interval joining P to the focus S .
- (ii) Show that, as P moves along the curve, the locus of M is a parabola, and state its focal length.
- (e) Find the value of the constants a and b if $x^2 + 3x - 4$ is a factor of the polynomial $P(x) = x^3 + x^2 + ax + b$. 2
- (f) The point $P(3, 3\frac{1}{2})$ divides the interval AB externally in ratio 3:1. If B is $(1, 3)$, find the co-ordinates of A . 2
- (g) Consider the cubic curve $y = x^3 - x$. 4
- (i) Write down the equation of the tangent to the curve at the point where $x = a$, in terms of a .
- (ii) Find, in terms of a , the x co-ordinate of the point where this tangent meets the curve again.

This is the end of the paper.

QUESTION 1

(a) $-\frac{\sqrt{3}}{2}$

(b)(i) $2-x \leq 3$ or $-2+x \leq 3$

$-x \leq 1$ $x \leq 5$

$-1 \leq x \leq 5$

(ii) $(x+2)(x-2) < 0$

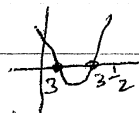
$-2 < x < 2$ ~~$x < 2$~~

(iii) $2x-6 < 4x^2-24x+36$

$4x^2-26x+42 > 0$

$2x^2-13x+21 > 0$

$(2x-7)(x-3) > 0$

$x < 3$ or $x > 3\frac{1}{2}$ 

(c) $P(3) = 27-15-1 = 11$

$R = 11$

(e) $2 \frac{(2t)}{1+t^2} + \frac{1-t^2}{1+t^2} = 0$

$4t + 1 - t^2 = 0$

$t^2 - 4t - 1 = 0$

$t = \frac{4 \pm \sqrt{20}}{2}$

$\tan \frac{\theta}{2} = 2 + \sqrt{5}$ or $2 - \sqrt{5}$

$\frac{\theta}{2} = 76.717^\circ, -13.282^\circ$

$\theta = 153.434^\circ$ or -26.566°

(f) $\cos 30 = \cos(20 + 10)$

$= \cos 20 \cos 10 - \sin 20 \sin 10$

$= (2\cos^2 10 - 1) \cos 10 - 2 \sin 10 \cos 10 \sin 10$

$= 2\cos^3 10 - \cos 10 - 2(1 - \cos^2 10) \cos 10$

$= 2\cos^3 10 - \cos 10 - 2\cos 10 + 2\cos^3 10$

$= 4\cos^3 10 - 3\cos 10$

(d) $\cos(\theta + \frac{\pi}{6}) = -\frac{1}{2}$

$\theta + \frac{\pi}{6} = 120, 360+120, 360+120, 2 \times 360+120$

$-120, -360+120, -360-120, 2 \times 360+120$

$\theta + \frac{\pi}{6} = 2n\pi + \frac{2\pi}{3}$ or $2n\pi - \frac{2\pi}{3}$

$\theta = 2n\pi + \frac{\pi}{2}$ or $2n\pi - 5\frac{\pi}{6}$

(g)(i) $y = x^{\frac{3}{2}} - x^{-1}$

$y' = \frac{3}{2} x^{\frac{1}{2}} - (-x^{-2})$
 $= \frac{3}{2} \sqrt{x} + \frac{1}{x^2}$

(ii) $(1-x^2)^4 \times 4(1+x^2)^3 \times 2x +$

$(1+x^2)^4 \times 4(1-x^2)^3 \times (-2x)$

$8x(1-x^2)^4(1+x^2)^3 - 8x(1+x^2)^4(1-x^2)^3$

(iii) $\frac{(1-x) \times 1 - (1+x) \times (-1)}{(1-x)^2}$

$\frac{2}{(1-x)^2}$

$\frac{2}{(1-x)^2}$

YR11 Yearly exam 2008 Continuers.
extension

$$\begin{aligned} 2) a) \cos x - \sin x &= \sqrt{2} \left(\frac{1}{\sqrt{2}} \cos x - \frac{1}{\sqrt{2}} \sin x \right) \\ R &= \sqrt{1^2 + (-1)^2} = \sqrt{2} \\ &= \sqrt{2} \end{aligned}$$

$$\begin{aligned} R &= \sqrt{2} \\ \text{and } \cos \alpha &= \frac{1}{\sqrt{2}} \\ \sin \alpha &= \frac{1}{\sqrt{2}} \end{aligned} \quad \left. \vphantom{\begin{aligned} R &= \sqrt{2} \\ \text{and } \cos \alpha &= \frac{1}{\sqrt{2}} \\ \sin \alpha &= \frac{1}{\sqrt{2}} \end{aligned}} \right\} \alpha = 45^\circ$$

$$\text{So } \cos x - \sin x = \sqrt{2} \cos(x + 45^\circ).$$

$$b) (i) \text{ RHS } \frac{1}{1 - \sin \theta} - \frac{1}{1 + \sin \theta}$$

$$\frac{1 + \sin \theta - (1 - \sin \theta)}{(1 - \sin \theta)(1 + \sin \theta)}$$

$$= \frac{2 \sin \theta}{1 - \sin^2 \theta}$$

$$= \frac{2 \sin \theta}{\cos^2 \theta}$$

$$= \frac{2 \sin \theta}{\cos \theta \cos \theta}$$

$$= 2 \tan \theta \sec \theta = \text{LHS}$$

$$\text{Using } S^2 + C^2 = 1$$

(2)

$$(b) (ii) \frac{\sin 3x}{\sin x} - \frac{\cos 3x}{\cos x}$$

$$= \frac{\sin 3x \cos x - \cos 3x \sin x}{\sin x \cos x}$$

$$= \frac{\sin(3x - x)}{\frac{1}{2} \sin 2x}$$

$$= \frac{2 \sin 2x}{\sin 2x} = 2$$

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

(2)

$$(c) \quad 3x^3 - 6x^2 + 3x + 1 = 0$$

$$\alpha + \beta + \gamma = -\frac{b}{a} = -\frac{-6}{3} = 2$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a} = \frac{3}{3} = 1$$

$$\alpha\beta\gamma = -\frac{d}{a} = -\frac{1}{3}$$

$$(i) \alpha\beta\gamma = -\frac{1}{3} \quad (1)$$

$$(ii) \alpha\beta + \alpha\gamma + \beta\gamma = 1 \quad (1)$$

$$(iii) \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\alpha\beta + \alpha\gamma + \beta\gamma}{\alpha\beta\gamma} = \frac{1}{-\frac{1}{3}} = -3 \quad (2)$$

d (i) $p(x) = x^3 + 6x^2 - x - 30$

$$P(2) = 8 + 24 - 2 - 30 = 32 - 32 = 0$$

$(x-2)$ is a factor.

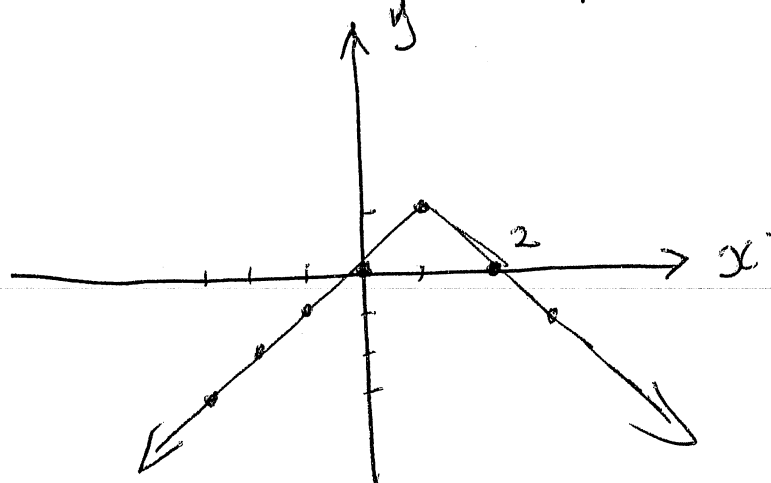
Could also have: $(x+3)$ or $(x+5)$.

$$\begin{array}{r} x^2 + 8x + 15 \\ (ii) \quad x-2 \overline{) x^3 + 6x^2 - x - 30} \\ \underline{x^3 - 2x^2} \\ 8x^2 - x - 30 \\ \underline{8x^2 - 16x} \\ 15x - 30 \\ \underline{15x - 30} \\ 0 \end{array}$$

$$P(x) = (x-2)(x^2+8x+15)$$
$$= (x-2)(x+3)(x+5)$$

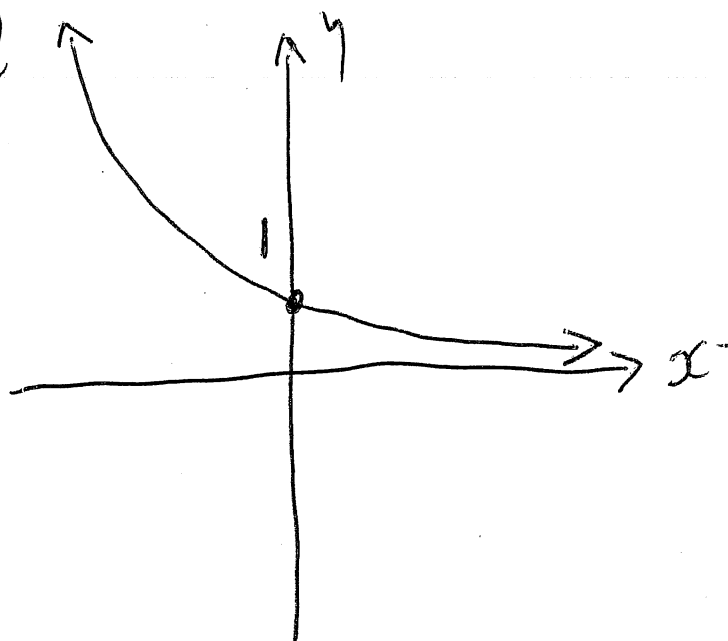
2

(e) (i) $y = 1 - |x - 1|$



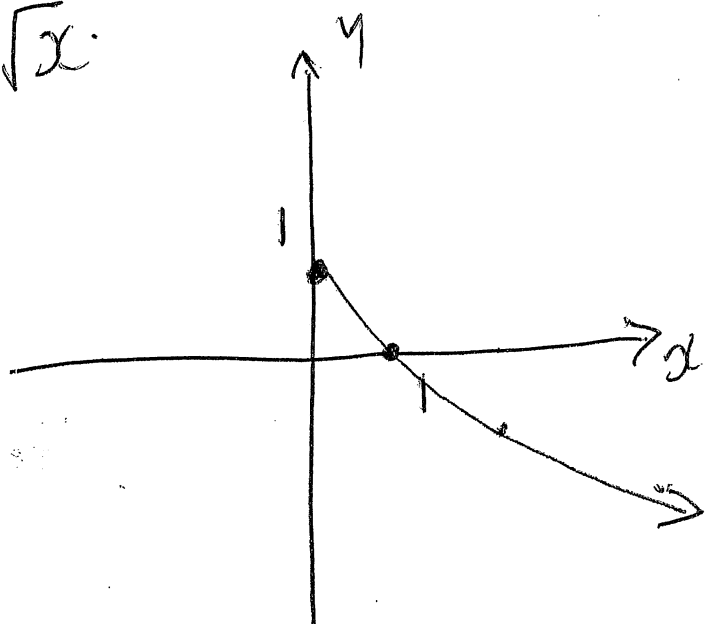
(2)

(ii) $y = 2^{-x}$



(2)

(iii) $y = 1 - \sqrt{x}$



(2)

(f)

$$\sin 2x = \sin x$$

radians

$$2 \sin x \cos x - \sin x = 0$$

$$\sin x (2 \cos x - 1) = 0$$

$$\sin x = 0$$

$$2 \cos x - 1 = 0 \Rightarrow \cos x = \frac{1}{2}$$

General solns are:

$$\theta = \pi n + (-1)^n \times 0 \quad (2) \quad n \text{ integer}$$

$$\text{and } \theta = 2\pi n \pm \frac{\pi}{3} \quad (2) \quad n \text{ integer.}$$

Question 3

i) 4

$$\left(x + \frac{1}{x}\right)^2 - 6\left(x + \frac{1}{x}\right) + 8 = 0$$

$$\text{let } u = x + \frac{1}{x}$$

$$\therefore u^2 - 6u + 8 = 0$$

$$(u-4)(u-2) = 0 \quad (1)$$

$$u=4 \text{ or } u=2 \quad (1)$$

$$x + \frac{1}{x} = 4 \text{ or } x + \frac{1}{x} = 2$$

$$x^2 - 4x + 1 = 0 \text{ or } x^2 - 2x + 1 = 0 \quad (1)$$

$$x = 2 \pm \sqrt{3} \text{ or } x = 1 \quad (1)$$

b) $y = \frac{1}{3}x - 2$ and $y = -2x - 5$

where $m_1 = \frac{1}{3}$ $m_2 = -2$

Acute angle $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$

$$= \left| \frac{\frac{1}{3} - (-2)}{1 + \frac{1}{3}(-2)} \right|$$

$$\tan \theta = 7$$

$$\theta = 81.52^\circ$$

d) $P(2ap, ap^2)$ $S(0, a)$

(i) Coords of M are

$$\left(\frac{2ap + 0}{2}, \frac{ap^2 + a}{2} \right)$$

ii $M\left(ap, \frac{a}{2}(p^2 + 1)\right)$ (1)

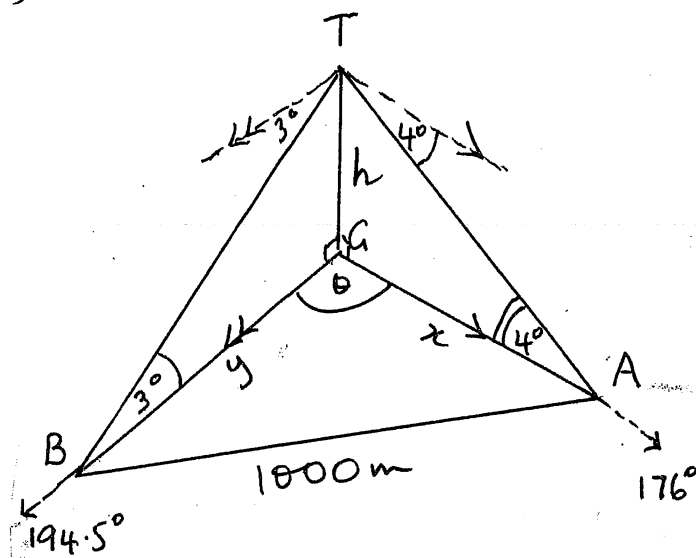
(ii) let $x = ap$, $y = \frac{a}{2}(p^2 + 1)$

Since $p = \frac{x}{a} \Rightarrow y = \frac{a}{2}\left(\frac{x^2}{a^2} + 1\right)$

or $x^2 = 2a\left(y - \frac{a}{2}\right)$

Focal length is $\frac{a}{2}$ units

(c) 5



(i) $\angle AGB = \theta = 18.5^\circ$ (1)

(ii) In $\triangle AGT$, $\tan 4^\circ = \frac{h}{x}$ (1/2)

In $\triangle BGT$, $\tan 3^\circ = \frac{h}{y}$ (1/2)

In $\triangle ABG$ by COSINE RULE

$$1000^2 = x^2 + y^2 - 2xy \cos 18.5^\circ$$

(iii) $x = h \cot 4^\circ$, $y = h \cot 3^\circ$

$$1000^2 = h^2 \cot^2 4^\circ + h^2 \cot^2 3^\circ - 2h^2 \cot 4^\circ \cot 3^\circ \cos 18.5^\circ$$

$$h^2 = \frac{1000^2}{[\cot^2 4^\circ + \cot^2 3^\circ - 2 \cot 4^\circ \cot 3^\circ \cos 18.5^\circ]}$$

$\therefore h \approx 140m$ (1)

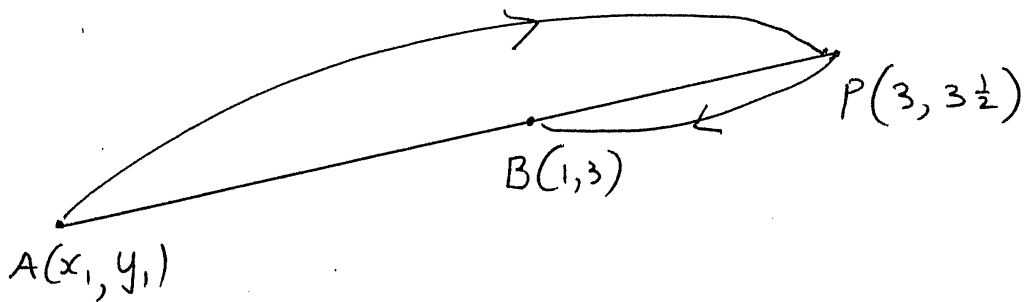
(e) $x^2 + 3x - 4 = (x+4)(x-1)$ $\therefore$ If a factor then $(x+4)$ and $(x-1)$ are factors.

(f) $P(-4) = -64 + 16 - 4a + b = 0$

$P(1) = 1 + 1 + a + b = 0$

$$\Rightarrow \begin{cases} -4a + b = 48 & \text{--- (1)} \\ a + b = -2 & \text{--- (2)} \end{cases} \Rightarrow \begin{cases} a = -10 \\ b = 8 \end{cases}$$

(f) 2



let
AP:PB = -3:1
(external division)

$$\left[\frac{1(x_1) + -3(1)}{-3+1}, \frac{1(y_1) + -3(3)}{-3+1} \right] = \left[3, 3\frac{1}{2} \right] \quad \text{--- (1)}$$

$$\frac{x_1 - 3}{-2} = 3 \Rightarrow x_1 = -3 \quad \text{and} \quad \frac{y_1 - 9}{-2} = 3\frac{1}{2} \Rightarrow y_1 = 2$$

$\therefore$ Coords of A $(-3, 2)$ --- (1)

(g) $y = x^3 - x$ --- (C)

(i) grad. tangent $\frac{dy}{dx} = 3x^2 - 1$ and when $x = a$ grad = $3a^2 - 1$ --- (1)

$\therefore$ Eqⁿ tangent is $y - (a^3 - a) = (3a^2 - 1)(x - a)$ --- (2)

or $y = x[3a^2 - 1] - 2a^3$ --- (D) --- (1)

(ii) Solving (C) and (D) $\Rightarrow x^3 - x = x[3a^2 - 1] - 2a^3$

ie $x^3 - 3a^2x + 2a^3 = 0$ (*)
 $x = a$ is a double root of this equation (since tangent at $x = a$)

$\therefore$ sum of roots of (*) is

$$a + a + \gamma = -\frac{(-3a^2)}{1} = 3a^2 \quad \text{--- (2)}$$

$$\gamma = 3a^2 - 2a$$

$\therefore$ This is the x coord. of other pt where tangent intersects curve again.